Epistemic Social Learning: Scalable Latent Opponent Modeling under Endogenous Interaction

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Abstract

When agents interact in multi-agent environments, they must adapt to diverse and previously unseen opponents under limited observations. Modeling opponent behavior can improve adaptation, but existing approaches often scale poorly with population size, rely on pairwise modeling, or assume stationary and exogenous data distributions. Moreover, many methods handle only specific classes of opponents, such as fixed or learning agents, but not both within a unified framework. We propose Epistemic Social Learning (ESL), a framework that enables agents to simultaneously infer latent behavioral structure and adapt their strategies in heterogeneous, non-stationary environments. ESL decomposes learning into two coupled processes: (i) interaction-level belief updates over a shared set of latent behavioral prototypes, and (ii) population-level prototype learning from belief-weighted interaction data. This induces an endogenous feedback loop in which beliefs shape policies, policies generate data, and data updates both beliefs and shared representations. By representing interactions through a small set of shared prototypes, ESL reduces modeling complexity from $O(N^2)$ pairwise relationships to $O(K)$ latent types, enabling scalable and interpretable adaptation. We formulate ESL as a stochastic approximation scheme with controlled Markov noise and show that its dynamics track a differential inclusion characterized by invariant measures of the fast belief process. Empirically, ESL consistently improves adaptation and recovers latent behavioral structure across repeated-game environments with fixed, adaptive, and belief-conditioned opponents.

1 Introduction

Strategic interaction in repeated multi-agent environments requires agents to reason about the behavior of others under uncertainty. Classical game-theoretic models assume that agents possess detailed knowledge of opponents' utilities or can compute equilibrium strategies through backward induction [17]. In practice, however, agents typically interact with unfamiliar opponents and must infer their behavioral tendencies from limited observations.

A central challenge in such settings is *scalability under heterogeneity*. In a population of N agents, naive opponent modeling requires learning distinct models for each interaction, leading to $O(N^2)$ pairwise dependencies. While Bayesian opponent modeling is effective in small-scale settings, it becomes intractable in large populations, as it requires maintaining separate posterior beliefs for each opponent [18]. Existing approaches in multi-agent reinforcement learning (MARL) mitigate this either by treating other agents as part of a non-stationary environment or by learning centralized joint representations [38]. Other approaches explicitly model opponent learning dynamics, but often rely on gradient access to opponents or scale poorly with population size. These methods typically sacrifice interpretability, fail to exploit shared behavioral structure, or struggle to generalize across heterogeneous and evolving populations.

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A natural hypothesis in many environments is that agents can be described by a small number of latent behavioral types. Evidence for such structure appears across evolutionary dynamics [12], agent-based social models [5], and tag-based cooperation mechanisms [20, 33]. However, existing approaches either assume these latent types are known a priori, infer them only at the population level without integrating them into agents' decision-making, or rely on offline clustering that fails under non-stationary interactions [27].

This raises a fundamental question:

> *Can agents simultaneously infer latent behavioral structure and adapt their strategies in a scalable and principled way under endogenous, non-stationary interaction dynamics?*

To address this, we introduce the *Epistemic Social Learner* (ESL), a framework for *online latent-structure learning in multi-agent systems*. ESL decomposes learning into two coupled processes: (i) *micro-level belief updates*, where agents maintain Bayesian beliefs over a finite set of latent behavioral prototypes, and (ii) *macro-level prototype learning*, where shared behavioral structure is learned from population-level interaction data.

At a structural level, ESL resembles a soft-assignment procedure: beliefs act as probabilistic responsibilities over latent prototypes, and parameters are updated accordingly. Crucially, however, the data-generating process is *endogenous*: agents act based on beliefs, interactions generate observations, and these observations in turn update both beliefs and prototypes. This induces a feedback loop in which the data distribution evolves with the model itself, fundamentally departing from classical latent-variable settings such as EM.

This perspective yields a key advantage: instead of modeling $O(N^2)$ pairwise interactions, ESL learns a shared K-dimensional latent behavioral space. Individual interactions are mediated through K-dimensional belief vectors, reducing modeling complexity from quadratic to $O(K)$. For example, with $N = 100$ agents and $K = 5$ latent types, this replaces 9900 pairwise relationships with just 5 shared prototypes. This reduction preserves expressivity while enabling scalable and interpretable reasoning about heterogeneous populations.

On the theoretical side, we formulate ESL as a stochastic approximation process with controlled Markov noise induced by belief-dependent interaction dynamics. Despite the endogenous non-stationarity, we show that the iterates track an averaged mean-field dynamics characterized by invariant measures of the fast belief process, and that their limiting behavior is described by internally chain transitive (ICT) sets of a differential inclusion under standard stability conditions.

Contributions

- **Epistemic social learning framework.** We introduce ESL, a two-timescale architecture that jointly learns latent behavioral prototypes (macro structure) and interaction-specific epistemic beliefs (micro inference), enabling adaptive reasoning in heterogeneous multi-agent systems without requiring equilibrium computation or offline data.

- **Scalable latent representation.** ESL replaces $O(N^2)$ pairwise opponent modeling with $O(K)$ shared prototypes, providing a compact and interpretable representation of population behavior that generalizes across agents and interactions.

- **Endogenous stochastic approximation analysis.** We formulate ESL as a stochastic approximation scheme with controlled Markov noise arising from belief-dependent data generation, extending classical analysis to feedback-coupled multi-agent settings.

- **Differential inclusion characterization.** We show that the learning dynamics track a set-valued mean-field differential inclusion, with limit sets characterized as internally chain transitive (ICT) sets induced by invariant regimes of the fast belief process.

- **Empirical validation.** We demonstrate that ESL recovers latent behavioral structure and improves adaptation in repeated games with heterogeneous and evolving opponents.

2 Related Work

Latent Variable Models and Strategy Clustering. Latent-variable approaches for modeling heterogeneous behavior are well-established in expectation-maximization (EM) frameworks. Recent extensions in multi-agent learning employ variational autoencoders (VAEs) to learn latent strategy spaces from offline trajectories, followed by clustering to guide downstream decision-making [27]. Similarly, HMM-based approaches [19] impose structured latent dynamics via predefined transition models.

A common limitation of these approaches is that representation learning is performed on exogenous, fixed datasets and remains decoupled from the strategic interaction process. Consequently, learned structures are brittle under distributional shift when agents adapt online. In contrast, ESL operates in a fully endogenous setting, where observations arise from belief-dependent interactions and the latent structure is continuously updated.

Bayesian Theory-of-Mind and Opponent Modeling. A large body of work in both classical and deep multi-agent systems assumes agents maintain beliefs over opponent types using Bayesian inference, including cognitive hierarchy models [10, 2, 31] and imperfect-information game solvers [18]. Early deep opponent modeling approaches such as DRON and DPIQN [21, 23] incorporate auxiliary networks to predict opponent behavior and condition policy updates.

However, these methods typically maintain independent models for each opponent, leading to quadratic $O(N^2)$ scaling in population settings, and often rely on stationarity or equilibrium-based priors. More recent open-ended and in-context opponent modeling approaches [24, 25], as well as emerging foundation-model-based agent modeling paradigms, adapt online without fixed parametric assumptions, but do not impose shared latent structure across agents. ESL addresses this limitation by projecting agents into a shared K-dimensional latent space, enabling scalable belief-based inference while coupling representation learning with interaction dynamics.

Multi-Agent Reinforcement Learning and Opponent Shaping. In deep multi-agent reinforcement learning, methods such as LOLA [14] explicitly differentiate through opponent updates to shape their learning dynamics, but require white-box access to opponent parameters and scale poorly in heterogeneous settings. Recursive reasoning approaches such as PR2 [35] and GR2 [36] extend reinforcement learning to model higher-order beliefs, but typically rely on centralized training and fixed agent populations.

Meta-learning approaches, such as ToMnet [32], learn behavioral embeddings from trajectories to enable rapid adaptation, while recent work shows opponent representations can emerge implicitly in recurrent agents [30]. However, these representations are typically black-box and entangled with policy networks.

ESL differs by explicitly disentangling belief inference from policy execution, enforcing an interpretable structure via Bayesian posteriors over discrete latent prototypes while operating in a black-box interaction setting.

Learning in Games and Equilibrium Concepts. Classical game-theoretic models often assume agents hold correct beliefs about their opponents. To relax this assumption, Self-Confirming Equilibrium (SCE) [15] requires beliefs to be correct only along the realized path of play, while Quantal Response Equilibrium (QRE) [29] models stochastic, boundedly rational responses. Prior work has studied learning dynamics that converge to SCE under various assumptions [16].

ESL is related in that agents act based on belief-induced models and stochastic best responses. The resulting dynamics give rise to invariant regimes characterized by internally chain transitive sets, which provide a dynamical systems perspective on such behavioral consistency. Unlike classical settings, ESL operates in a latent representation space with endogenous data generation.

Stochastic Approximation under Controlled Markov Noise. The analysis of ESL builds on stochastic approximation with controlled Markov noise [7] and the theory of asymptotic pseudotrajectories [3, 4]. Unlike standard two-timescale settings where both processes evolve simultaneously, ESL employs a frozen-batch update scheme, allowing the fast belief dynamics to evolve under a

stationary kernel within each batch. This enables the application of existing controlled Markov noise results to characterize the limiting behavior via differential inclusions.

Positioning. Existing approaches typically address latent structure learning, Bayesian inference, or strategic adaptation in isolation, and assume that the data distribution is externally given or stationary. ESL departs from this paradigm by treating the data distribution as an outcome of belief-dependent interaction, coupling representation learning, inference, and policy adaptation within a unified stochastic approximation framework.

3 Preliminaries

We consider a population of N agents indexed by $[N] = \{1, \dots, N\}$, sharing a common finite action set $\mathcal{A}$. The environment unfolds over discrete rounds $t = 0, 1, 2, \dots$. At each round, a number of interactions $L_t \in \{L_{\min}, \dots, L_{\max}\}$, with $1 \le L_{\min} \le L_{\max} < \infty$, is drawn, and a set $\mathcal{E}_t \subset \{(i,j) \in [N] \times [N] : i \neq j\}$ of L_t ordered pairs is sampled uniformly without replacement, from all possible pairs, so that only a subset of interactions occurs in each round. Each matched pair $(i,j) \in \mathcal{E}_t$ selects actions $a_i^t, a_j^t \in \mathcal{A}$ and receives payoffs $u_i(a_i^t, a_j^t)$ and $u_j(a_j^t, a_i^t)$ from a two-player stage game $u : \mathcal{A} \times \mathcal{A} \to \mathbb{R}$. Interactions within a round are treated as simultaneous, and all resulting observations are aggregated before belief updates are applied at the end of the round.

Agents observe opponents through a stochastic signaling mechanism. For each $(i,j) \in \mathcal{E}_t$, an indicator $W_{ij,t} \in \{0,1\}$ with $\mathbb{P}(W_{ij,t} = 1) > 0$ determines whether agent i receives a signal $s_{ij,t} \sim P(\cdot \mid a_j^t)$ about agent j. In the primary setting, we consider noiseless observations where $s_{ij,t} = a_j^t$, while the general formulation allows for stochastic signals.

We assume the signal model is sufficiently informative such that distinct latent behavioral prototypes induce distinguishable distributions over signals. A formal identifiability condition is stated in Assumption 1 (Section 4.4). We use the term *prototypes* to denote learned, parameterized behavioral representations, distinguishing them from fixed latent *types* in classical models, as prototypes are adapted over time through interaction.

We assume there exist K latent behavioral prototypes indexed by $k \in [K]$, each parameterized by $\theta_k \in \Theta \subset \mathbb{R}^{|\mathcal{A}|}$, defining a policy over actions via a softmax parameterization $\pi_k(a \mid \theta_k) = \mathrm{softmax}(\theta_k)[a]$. The induced signal likelihood for prototype k is

$$L_k(s \mid \theta_k) \;:=\; \sum_{a \in \mathcal{A}} P(s \mid a)\, \pi_k(a \mid \theta_k) \tag{1}$$

which reduces to $L_k(a \mid \theta_k) = \pi_k(a \mid \theta_k)$ in the primary setting. We denote the collection of prototype parameters by $\Theta_t = \{\theta_k\}_{k \in [K]}$. The likelihood $L_k(s \mid \theta_k)$ thus represents the probability of observing signal s under prototype k, induced by its action distribution and the observation model. In the noiseless setting, this reduces to the probability of the observed action itself, while in the general case it captures the stochastic mapping from actions to signals.

Each agent i maintains, for every opponent $j \neq i$, a belief vector $b_{i \to j,t} \in \Delta_K^\delta$, where

$$\Delta_K^\delta \;:=\; \left\{ b \in \mathbb{R}^K : \sum_{k=1}^K b_k = 1,\; b_k \ge \delta \right\} \tag{2}$$

for some $\delta > 0$ with $\delta < 1/K$. The component $b_{i \to j,t}[k]$ represents the probability that agent j belongs to prototype k; projection onto Δ_K^δ ensures strictly positive beliefs and prevents degeneracy in long-run interactions. This projection acts as a technical regularization that keeps the global belief state within a compact convex set, which is important for stability of the coupled learning dynamics and for the analysis in Section 4.4. The global belief state is $B_t := \{b_{i \to j,t}\}_{i \neq j}$.

When agent i interacts with agent j at round t, it forms a predictive distribution over j's actions by marginalizing over prototypes, $\hat{\pi}_{j|i,t}(a_j) = \sum_{k=1}^K b_{i \to j,t}[k]\, \pi_k(a_j \mid \theta_k)$, and selects actions according to a logit best-response policy with inverse temperature $\lambda > 0$:

$$\pi_i(a_i \mid b_{i \to j,t}, \Theta_t) \;=\; \frac{\exp\{\lambda\, U_i(a_i \mid b_{i \to j,t}, \Theta_t)\}}{\sum_{a' \in \mathcal{A}} \exp\{\lambda\, U_i(a' \mid b_{i \to j,t}, \Theta_t)\}} \tag{3}$$

where $U_i(a_i \mid b_{i\to j,t}, \Theta_t) = \sum_{a_j \in \mathcal{A}} \hat{\pi}_{j|i,t}(a_j)\, u_i(a_i, a_j)$ is the expected payoff of action a_i. This induces a feedback loop in which beliefs influence actions, actions generate observations, and observations drive updates to both beliefs B_t and prototype parameters Θ_t.

4 Method: Epistemic Social Learning

ESL models multi-agent interaction as a coupled two-timescale learning process driven by endogenous feedback. At the fast timescale, agents maintain pairwise epistemic beliefs over latent behavioral prototypes and update them online from observed interaction signals, yielding interaction-specific opponent models that adapt continuously over time. At the slower timescale, these belief-induced models are aggregated across interactions to learn a shared set of prototype parameters that capture recurring behavioral structure in the population.

The central challenge in ESL is the circularity of social learning: agent beliefs determine their policies, these policies generate the interaction signals, and these signals in turn drive the evolution of the shared behavioral prototypes. This feedback loop induces a self-consistent, non-stationary learning dynamic where the data distribution is not exogenous but emerges endogenously from the current model state. We resolve this circularity by decomposing the learning process into two coupled timescales: a fast process for interaction-specific belief-based inference and a slow process for representation adaptation. The overall ESL procedure is summarized in Algorithm 1.

4.1 Fast Timescale: Epistemic Inference under Endogenous Interaction

In interactive settings, the data distribution is shaped by the agents' own policies, which evolve as beliefs are updated over time. Consequently, models fitted to historical observations may become misaligned with the current interaction dynamics. Moreover, since each agent's behavior depends on its beliefs about others, interactions induce a feedback loop in which agents continuously adapt to one another, a phenomenon closely related to belief-based and recursive reasoning in multi-agent systems [9, 1].

To address this, ESL performs online belief-based inference at the level of individual interactions. Beliefs are updated using signals generated by the interaction process itself, yielding a streaming form of Bayesian filtering over latent behavioral types [13]. Because the update is followed by projection onto Δ_K^δ, it is more precisely a regularized variant of Bayesian filtering operating under a non-stationary, policy-dependent data distribution induced by the interaction dynamics.

Formally, for each observed interaction $(i, j) \in \mathcal{E}_t$ with $W_{ij,t} = 1$, agent i performs a projected Bayesian update:

$$\begin{aligned} \tilde{b}_{i\to j,t+1}[k] &\propto b_{i\to j,t}[k]\, L_k(s_{ij,t} \mid \theta_{k,t}), \\ b_{i\to j,t+1} &= \Pi_{\Delta_K^\delta}\big(\tilde{b}_{i\to j,t+1}\big) \end{aligned} \tag{4}$$

where the proportionality denotes normalization over k. Beliefs corresponding to unobserved interactions remain unchanged.

The projection $\Pi_{\Delta_K^\delta}$ ensures that beliefs remain strictly positive and prevents degeneracy in long-run interactions, while confining the belief dynamics to a compact convex set. This compactness is critical for stability of the coupled stochastic approximation and ensures that the score function $\nabla_{\theta_k} \log L_k$ remains well-defined and bounded [11, 8]. The resulting update can be viewed as a projected (regularized) variant of Bayesian filtering, where the projection enforces stability by preventing belief collapse at the boundary of the simplex.

The belief-weighted predictive distributions $\hat{\pi}_{j|i,t}$ induced by $b_{i\to j,t}$ define interaction-specific opponent models that evolve continuously as new observations are incorporated, connecting to belief-based opponent modeling approaches in multi-agent learning [22].

Over time, successive updates refine these models, implicitly capturing interaction effects arising from the mutual adaptation of agents.

The dynamics are stochastic due to random interaction sampling and partial observability, and their stability is governed by the informativeness of observations and the projection parameter δ, which controls a tradeoff between adaptivity and robustness.

Algorithm 1 Epistemic Social Learning (ESL)

Initialize: beliefs B_0, prototypes Θ_0, buffer $\mathcal{D} \leftarrow \emptyset$
for $t = 1, \ldots, T$ **do**
  Fast Timescale: Interaction and Belief Updates
  Sample interaction set $\mathcal{E}_t$
  for each $(i, j) \in \mathcal{E}_t$ **do**
    Sample actions $a_i^t \sim \pi_i(\cdot \mid b_{i\to j,t}, \Theta_t)$
    Observe signal $s_{ij,t}$ if $W_{ij,t} = 1$
    Update belief $b_{i\to j,t+1}$ via (4)
    Store $(i, j, s_{ij,t}, b_{i\to j,t})$ in $\mathcal{D}$
  end for
  Slow Timescale: Prototype Update
  if $t \bmod M = 0$ **then**
    Update Θ_t via (5) using $\mathcal{D}$
    Reset buffer $\mathcal{D} \leftarrow \emptyset$
  end if
end for

The slow-timescale updates are triggered every M interaction steps; we later index these updates by m to distinguish them from the fast interaction index t.

4.2 Slow Timescale: Prototype Adaptation under Endogenous Feedback

The belief-based inference process described above yields a collection of interaction-specific opponent models, each induced by the belief state at the time of observation. Since these models are formed under evolving interaction dynamics, any single belief-induced model may provide a partial or noisy explanation of behavior. To obtain a robust and transferable representation, ESL aggregates these models into a shared set of prototype parameters.

This aggregation operates at a slower timescale and integrates information across interactions through belief-weighted updates. Let $\mathcal{D}_m$ denote the set of interaction records collected over a batch of rounds, where each record consists of $(i, j, s, b^{\text{snap}})$ and b^{snap} is the belief at the time of observation. The prototype parameters are updated as

$$\begin{aligned} \bar{g}_{k,m} &= \frac{1}{|\mathcal{D}_m|} \sum_{(i,j,s,b^{\text{snap}})\in\mathcal{D}_m} b^{\text{snap}}[k]\, \nabla_{\theta_k} \log L_k(s \mid \theta_k), \\ \theta_k &\leftarrow \theta_k + \gamma_m \left(\bar{g}_{k,m} - \eta_{\text{reg}}\theta_k\right) \end{aligned} \tag{5}$$

where $\eta_{\text{reg}} > 0$ is an ℓ_2 regularization coefficient ensuring stability of the prototype parameters. Let m index the slow-timescale prototype updates, corresponding to interaction steps $t \in [mM, (m+1)M - 1]$, during which Θ_m remains fixed. The slow-timescale update bears structural resemblance to online EM [11]. However, a key distinction is that in ESL the weighting distribution b^{snap} is not computed from a fixed dataset, but arises endogenously from the interaction process. Since agent policies depend on beliefs, the data distribution itself evolves with the model state, violating the exogeneity assumptions underlying standard online EM analysis. This endogenous coupling necessitates the two-timescale formulation developed here. Within each batch, the prototype parameter Θ_m remains fixed, so the fast belief process evolves under a time-homogeneous Markov kernel P_{Θ_m} over the entire batch.

Consequently, mixing occurs under a stationary kernel within each batch, eliminating the need for Poisson equation analysis required in simultaneous-update stochastic approximation. The convergence analysis therefore reduces directly to the controlled Markov framework of [6].

Unlike classical latent-variable estimation where data samples are treated as exogenous, the update direction in ESL is driven by a controlled Markovian noise process. Because both the weights b^{snap} and the observed signals s depend on the evolving global belief state B_t, the noise in the gradient estimate is state-dependent. The slow-timescale update therefore aggregates information across the invariant regimes of the fast belief dynamics rather than over an i.i.d. sample stream.

Each observation contributes to all prototypes, weighted by the belief that the corresponding prototype explains the observed behavior. In this sense, the update aggregates competing behavioral hypotheses induced by the epistemic states across interactions, forming a shared representation that reflects their relative explanatory power.

This update defines a stochastic approximation scheme over a non-stationary data stream [7], where the effective data distribution is shaped by the evolving belief and policy dynamics.

In practice, aggregation is performed over batches of interaction data collected across multiple rounds, allowing the prototype parameters to integrate information from diverse interaction contexts while smoothing high-variance updates from individual interactions. The resulting slow-timescale dynamics provide a stable yet adaptive representation that tracks evolving behavioral structure in the population.

Remark (Staleness and averaging). The use of buffered beliefs b^{snap} introduces a delay between the time of observation and the prototype update. Since Θ_m is frozen within batch $\mathcal{D}_m$, this delay manifests as a discrepancy between the finite-batch empirical update and the invariant-measure average associated with the frozen fast process. In the appendix, this discrepancy is captured by a Markovian tracking bias term β_m, whose asymptotic vanishing is established via ergodic averaging for controlled Markov processes under timescale separation [6]. Thus, larger batch windows improve averaging of the fast dynamics, while smaller batch windows yield more responsive but noisier prototype adaptation.

In our setting, persistent stochastic interaction (with $\mathbb{P}(W_{ij,t} = 1) > 0$) and the strict positivity enforced by Δ_K^δ promote recurrent exploration of the fast state space. We treat this as part of the regularity regime under which the controlled-Markov averaging argument applies.

4.3 Two-Timescale Dynamical System Formulation

ESL induces a coupled two-timescale stochastic dynamical system [7, 26]. The fast process corresponds to belief updates, while the slow process governs prototype adaptation.

For fixed prototype parameters Θ, the belief state $B_t = \{b_{i\to j,t}\}_{i\neq j}$ evolves according to the projected Bayesian update in (4). Since interaction sampling, policies, and signal generation depend only on the current state (B_t, Θ), this defines a time-homogeneous Markov process on the compact state space

$$\mathcal{B} = (\Delta_K^\delta)^{N(N-1)}.$$

We analyze this coupled system under a quasi-static approximation. Since prototype parameters evolve on the slower update scale γ_m, they are treated as effectively constant relative to the fast belief dynamics. For fixed Θ, the fast process is therefore characterized through its invariant regimes, which induce the set-valued mean-field map $\mathcal{G}(\Theta)$ used in the next section.

On the slower timescale, prototype parameters are updated using stochastic gradients based on interaction data, as defined in (5). Let

$$H_k(B, i, j, s; \Theta) = b_{i\to j}[k]\, \nabla_{\theta_k} \log L_k(s \mid \theta_k) - \eta_{\text{reg}}\theta_k \tag{6}$$

denote the instantaneous update contribution.

The resulting recursion can be written as a stochastic approximation scheme with controlled Markov noise:

$$\Theta_{m+1} = \Theta_m + \gamma_m\, \widehat{H}(\mathcal{D}_m, \Theta_m), \tag{7}$$

where $\widehat{H}$ is the empirical average of H_k over a batch of interactions collected while Θ_m is fixed. In particular, the randomness in the slow update is induced by the fast belief process rather than by an exogenous i.i.d. data source.

This formulation highlights the separation of timescales: belief updates occur at each interaction, while prototype parameters evolve more slowly based on aggregated data. The coupling between these processes induces a non-stationary, policy-dependent learning dynamic, whose limiting behavior can be characterized via differential inclusion techniques [3].

Under standard stochastic approximation arguments, the limiting behavior of the slow-timescale iterates is governed by averaged update directions induced by invariant regimes of the fast belief

process. In particular, this leads to a set-valued mean-field map of the form

$$\mathcal{G}(\Theta) = \{g_\mu(\Theta) : \mu \in \mathcal{I}(\Theta)\}, \tag{8}$$

where $\mathcal{I}(\Theta)$ denotes the set of invariant probability measures of the fast belief dynamics. Since the fast process has no auxiliary control variable beyond the frozen parameter Θ, the relevant ergodic occupation measures reduce here to invariant probability measures on $\mathcal{B}$ [6]. This characterization underlies the differential inclusion analyzed in the next section.

Proposition 1 (Frozen-batch simplification of controlled Markov noise)**.** *Within each batch, the prototype parameter Θ_m remains fixed, and the fast belief process evolves under a time-homogeneous Markov kernel P_{Θ_m}.*

As a result, mixing occurs under a stationary kernel within each batch, eliminating the need for Poisson equation analysis required in simultaneous-update stochastic approximation. The convergence analysis therefore reduces to the controlled Markov framework of [6].

4.4 Theoretical Analysis: Convergence under Endogenous Feedback

To formalize the convergence properties of the slow-timescale prototype adaptation under endogenous, belief-coupled non-stationarity, we analyze the stochastic approximation scheme as a discrete-time tracking of a continuous-time differential inclusion.

Let $\{\mathcal{F}_m\}_{m\geq 0}$ denote the natural filtration generated by the history of prototypes, beliefs, and interactions up to slow-time index m. Starting from the empirical recursion (7), we decompose the batch update into an averaged drift term, a martingale fluctuation term, and a residual tracking bias. Under the quasi-static approximation, this decomposition follows from adding and subtracting the conditional expectation of the empirical update (see Appendix A for details), yielding

$$\Theta_{m+1} = \Theta_m + \gamma_m\big(\xi_m + M_{m+1} + \beta_m\big), \tag{9}$$

where $\xi_m \in \mathcal{G}(\Theta_m)$ is a measurable selection of the set-valued mean-field map $\mathcal{G}$, M_{m+1} is a martingale difference noise term capturing finite-sampling fluctuations, and β_m is a Markovian tracking bias arising from the fact that the fast belief process does not fully equilibrate to an invariant regime over a finite batch.

To ensure that the mean-field map is well-behaved, we impose the following structural assumptions.

Assumption 1 (Local signal identifiability and smoothness)**.** *For each prototype k, the likelihood $L_k(s \mid \theta_k)$ is continuously differentiable in θ_k, and the score function $\nabla_{\theta_k} \log L_k(s \mid \theta_k)$ is Lipschitz continuous on Θ. Moreover, there exists an open set $\mathcal{N} \subset \Theta$ and a constant $\epsilon > 0$ such that for any distinct $k, k' \in [K]$,*

$$\inf_{(\theta,\theta')\in\mathcal{N}\times\mathcal{N}} \big\| L_k(\cdot \mid \theta) - L_{k'}(\cdot \mid \theta') \big\|_{\mathrm{TV}} \geq \epsilon.$$

Assumption 1 ensures that, in the regime where the dynamics concentrate, distinct prototypes induce distinguishable signal distributions, preventing collapse and maintaining informative gradient signals for the slow-timescale updates.

Assumption 2 (Step-size conditions)**.** *The sequence $\{\gamma_m\}$ satisfies*

$$\sum_{m=0}^{\infty} \gamma_m = \infty, \qquad \sum_{m=0}^{\infty} \gamma_m^2 < \infty,$$

and is eventually non-increasing.

Assumption 3 (Conditional second-moment bound)**.** *The martingale difference sequence $\{M_{m+1}\}$ satisfies*

$$\mathbb{E}\big[\|M_{m+1}\|^2 \mid \mathcal{F}_m\big] \leq C\big(1 + \|\Theta_m\|^2\big)$$

for some constant $C < \infty$.

Assumption 4 (Regularity of the mean-field map)**.** *For each fixed Θ, the fast-timescale belief process on $\mathcal{B}$ admits at least one invariant probability measure, so $\mathcal{I}(\Theta)$ is non-empty. The correspondence $\Theta \mapsto \mathcal{I}(\Theta)$ has non-empty compact values and is upper hemicontinuous. The induced set-valued mean-field map $\mathcal{G}(\Theta)$ is upper semicontinuous with non-empty, convex and compact values.*

Convexity of $\mathcal{G}(\Theta)$ follows from the convexity of $\mathcal{I}(\Theta)$ and the linearity of the averaging map (see Appendix A).

Lemma 1 (Boundedness of iterates)**.** *Under the update* (5) *with* $\eta_{\text{reg}} > 0$*, the sequence of iterates* $\{\Theta_m\}_{m\geq 0}$ *is almost surely bounded.*

Let $\Theta(\tau)$ denote the standard piecewise-linear interpolation of the iterates $\{\Theta_m\}$ with step sizes $\{\gamma_m\}$.

Theorem 1 (Asymptotic characterization of prototype dynamics)**.** *Under Assumptions 1, 2, 3, and 4, together with the boundedness established in Lemma 1, the interpolated process* $\Theta(\tau)$ *is almost surely an asymptotic pseudotrajectory (APT) of the differential inclusion*

$$\dot{\Theta} \in \mathcal{G}(\Theta). \tag{10}$$

Consequently, the limit set of the prototype parameter sequence $\{\Theta_m\}$ *is almost surely a compact internally chain transitive (ICT) set of* (10)*.*

Proof sketch. The proof maps the ESL recursion to the controlled Markov noise framework of [6] (Theorem 3.1), with asymptotic pseudotrajectory properties following from [4] (Theorem 3.5), and the ICT characterization from [3] (Theorem 5.7). Full details are provided in Appendix A.

Corollary 1 (Convergence to fixed points under additional conditions)**.** *If the invariant measure is unique for each* Θ*, so that the mean-field map* $\mathcal{G}(\Theta)$ *reduces to a single-valued vector field* $g(\Theta)$*, and the associated ODE*

$$\dot{\Theta}(\tau) = g(\Theta)$$

admits a unique globally asymptotically stable equilibrium Θ^**, then the iterates* $\{\Theta_m\}$ *converge almost surely to* Θ^**.*

5 Experiments

We evaluate the Epistemic Social Learner (ESL) along four complementary axes: (i) recovery of latent behavioral structure, (ii) strategic adaptation under heterogeneous and evolving interaction regimes, (iii) ablations of the key epistemic components, and (iv) empirical dynamical behavior relative to the theoretical analysis.

All experiments are conducted in repeated multi-agent interaction settings with endogenous data generation induced by belief-dependent policies. We consider cooperative, coordination, and adversarial regimes through standard repeated games. Across all experiments, we compare ESL with baselines spanning independent learning, opponent modeling, and opponent-shaping approaches. Methods use comparable policy parameterizations and matched computational budgets. Results are averaged over multiple random seeds and reported with mean and 95% confidence intervals. Additional implementation details are provided in Appendix A.

5.1 Experimental Setup and Evaluation Protocol

Tasks. We consider repeated two-player matrix games capturing distinct strategic regimes: Iterated Prisoner's Dilemma (IPD), Stag Hunt, and Matching Pennies. These environments represent cooperation under temptation, coordination under multiple equilibria, and adversarial interaction, respectively, while allowing precise control over latent behavioral structure.

Population generation. Agents are drawn from a mixture of K latent behavioral types. Each type corresponds to a prototype parameter $\theta_k^\star$, and individual agents exhibit stochastic deviations around their assigned prototype. This induces structured heterogeneity in the population while preserving a shared latent representation.

Interaction protocol. At each interaction round, pairs of agents are sampled and interact sequentially. Observations are aggregated into batches, and prototype parameters are updated at the batch level using belief-weighted stochastic gradients. The resulting data distribution is endogenous, as both interaction outcomes and future data depend on evolving beliefs and policies.

Population regimes. To evaluate robustness under varying levels of non-stationarity, we consider three regimes:

- **Fixed types:** agent policies remain stationary.
- **Adaptive agents:** agents update policies online via independent learning.
- **Belief-conditioned agents:** agents adapt policies based on internal models of their partners.

These regimes progressively increase strategic complexity and non-stationarity.

Baselines. We compare ESL against the following baselines:

- **Independent PPO:** [34] a standard policy-gradient method without opponent modeling.
- **M-FOS** [28]: a model-free opponent-shaping approach that accounts for long-term learning dynamics.
- **MBOM** [37]: a model-based opponent modeling method using imagined future responses.
- **Simple Opponent Model:** predicts opponent actions from recent interaction history and responds via best-response updates, without shared latent structure.
- **FCM/GMM Clustering:** applies static clustering to action statistics, ignoring endogenous interaction structure.

Evaluation metrics. We evaluate performance along three dimensions:

- **Task performance:** average return or win rate against held-out populations.
- **Prototype recovery:** matched cross-entropy (or KL divergence) between learned and ground-truth prototypes.
- **Belief accuracy:** cross-entropy and top-1 accuracy of inferred beliefs relative to true latent types.

5.2 Prototype Recovery and Strategic Adaptation

We evaluate ESL's ability to jointly infer latent behavioral structure and adapt strategies under heterogeneous and evolving interaction regimes.

Performance across regimes. Figure **??** shows task performance across fixed, adaptive, and belief-conditioned populations. Table **??** summarizes aggregate results.

[Placeholder: Performance vs population regimes]

[Placeholder: Aggregate performance table]

Prototype recovery. Figure **??** reports matched cross-entropy between learned and ground-truth prototypes over training.

[Placeholder: Prototype recovery curves]

Belief accuracy. Figure **??** reports belief prediction performance across regimes.

[Placeholder: Belief accuracy plots]

Analysis. Independent learning fails under heterogeneity due to lack of opponent modeling. Static clustering methods cannot recover meaningful structure under endogenous data. Opponent modeling baselines improve adaptation but do not capture shared latent structure. ESL consistently achieves both accurate prototype recovery and robust adaptation, with gains most pronounced in belief-conditioned regimes.

5.3 Scalability with Population Size

We evaluate the scalability of ESL as the number of agents increases while keeping the number of latent prototypes fixed.

Setup. We fix $K = 5$ and vary the population size $N \in \{10, 20, 50, 100, 200\}$.

Metrics. We measure (i) wall-clock time per update and (ii) task performance at convergence.

[Placeholder: Scaling plots showing compute vs N and performance vs N]

Analysis. Pairwise opponent modeling methods exhibit quadratic scaling in N, leading to increasing computational cost and degraded performance. In contrast, ESL maintains stable computational cost and performance across population sizes, consistent with its $O(K)$ latent representation.

5.4 Effect of Batch Size on Learning Dynamics

We evaluate the role of batch size M in stabilizing the coupled learning dynamics, directly testing the frozen-batch mechanism underlying Proposition 1.

Setup. We vary the batch size $M \in \{1, 5, 20, 50, 100, 500\}$ under the adaptive population regime, where non-stationarity is most pronounced.

Metrics. We report: (i) task performance at convergence, (ii) prototype recovery error, and (iii) convergence stability measured via variance of prototype updates.

[Placeholder: Batch size ablation plots]

Analysis. The results reveal a clear dependence of learning dynamics on batch size.

When $M = 1$, corresponding to simultaneous updates of beliefs and prototypes, learning becomes unstable: prototype updates exhibit high variance and fail to converge reliably. This reflects the dominance of state-dependent noise and tracking bias arising from insufficient mixing of the fast belief process.

As M increases, the dynamics stabilize. Intermediate batch sizes ($M \in [20, 100]$) achieve the best tradeoff between stability and responsiveness, yielding both low-variance updates and strong task performance.

For very large M, performance degrades due to staleness: prototype updates lag behind the evolving interaction dynamics, slowing adaptation.

These observations are consistent with the theoretical interpretation of Proposition 1: freezing prototype parameters within each batch allows the fast belief process to mix under a stationary kernel, reducing gradient noise and enabling stable stochastic approximation. The empirical results thus validate the role of timescale separation in controlling bias–variance tradeoffs in endogenous learning dynamics.

5.5 Self-Consistency and Dynamical Behavior

We evaluate whether ESL converges to self-consistent behavioral regimes, as predicted by the differential inclusion analysis in Section 4.

Self-consistency protocol. After training converges, we freeze the learned prototype parameters Θ and simulate the belief dynamics for an extended horizon to approximate the induced invariant distribution of the fast process.

We evaluate two quantities:

- **Gradient norm:** the norm of the belief-weighted update direction $\|\bar{g}(\Theta)\|$,
- **Self-confirmation gap:** the KL divergence between predicted opponent behavior $\hat{\pi}_{j|i}$ and empirical action distributions.

[Placeholder: Self-consistency plots]

Analysis. Across environments, the gradient norm decreases toward zero over training, indicating convergence to fixed points of the mean-field dynamics. Simultaneously, the self-confirmation gap diminishes, showing that the learned prototypes accurately predict the interaction data they induce.

This provides empirical evidence for convergence to self-consistent behavioral regimes, consistent with the characterization of fixed points as solutions satisfying $g_\mu(\Theta) = 0$.

Dynamical regimes. We further analyze prototype trajectories across environments.

In cooperative and coordination settings (IPD, Stag Hunt), trajectories converge toward stable configurations, consistent with fixed-point behavior.

In adversarial settings (Matching Pennies), trajectories exhibit persistent oscillations, with no convergence to a single point.

[Placeholder: Prototype trajectory plots]

Interpretation. The contrast between convergent and oscillatory regimes aligns with the differential inclusion framework: while some environments admit stable fixed points, others exhibit recurrent dynamics corresponding to more general internally chain transitive sets. These results empirically support the theoretical claim that ESL dynamics need not converge to a single equilibrium, but instead track invariant behavioral regimes induced by endogenous feedback.

6 Conclusion

Game-theoretic interpretation of limit sets. The internally chain transitive (ICT) sets identified by Theorem 1 admit a natural game-theoretic interpretation. Since agents act according to logit best-response policies (3) and update beliefs from realized interactions, the limiting objects of ESL need not coincide with Nash equilibria. Rather, they are better understood as *self-consistent behavioral regimes* in which beliefs, induced policies, and the resulting data distribution are mutually compatible under the endogenous learning dynamics.

Formally, the differential inclusion (10) governs the asymptotic evolution of the slow-timescale dynamics, and its internally chain transitive (ICT) sets characterize the invariant regimes of the system. Fixed points correspond to stationary self-consistent configurations, whereas more general ICT sets capture recurrent, non-equilibrium behavior induced by the endogenous belief–policy feedback loop.

This places ESL closer to bounded-rationality equilibrium notions than to exact strategic equilibrium. In particular, at a fixed point Θ^* of the differential inclusion, there exists an invariant measure $\mu^* \in \mathcal{I}(\Theta^*)$ such that

$$g_{\mu^*}(\Theta^*) = 0.$$

That is, the prototype parameters are stationary under the data distribution induced by μ^*. In this sense, Θ^* correctly describes the interaction patterns that it itself generates through beliefs and policies, which is the natural analogue of a *Self-Confirming Equilibrium* (SCE), where beliefs need only be accurate along the realized path of play [15].

At the same time, the logit best-response policy (3) induces the standard *Quantal Response Equilibrium* (QRE) structure of [29], so fixed points of ESL can be viewed as QRE-like responses to self-confirmed beliefs rather than to the true population distribution.

More complex ICT structures, such as limit cycles or recurrent sets, need not admit a standard equilibrium interpretation. Instead, they correspond to persistent behavioral oscillations in which no single prototype is self-confirming, but the system evolves through a recurring sequence of mutually consistent configurations. These objects capture the non-convergent dynamics that can arise naturally under endogenous belief–policy feedback.

Limitations and future directions. Several extensions remain open. First, while ESL achieves $O(K)$ scaling, the choice of K remains a modeling assumption; adaptive or nonparametric extensions could further improve flexibility. Second, the current framework assumes discrete latent prototypes, and extending to continuous or hierarchical representations is a promising direction. Third, while the frozen-batch scheme enables clean theoretical analysis, understanding fully online variants without

explicit timescale separation remains an important challenge. Finally, scaling ESL to large, high-dimensional environments and integrating richer function approximators are key steps toward broader applicability.

Overall, ESL provides a unified framework for reasoning, representation, and adaptation in multi-agent systems with endogenous data generation, offering both theoretical guarantees and a scalable alternative to existing opponent modeling approaches.

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A Appendix: Convergence Analysis under Controlled Markov Noise

This appendix establishes Theorem 1 using stochastic approximation with controlled Markov noise [6, 7] and asymptotic pseudotrajectory analysis [3, 4]. The recursion indexed by m corresponds to the iteration index in [6], with each update operating on batch-level observations. The analysis proceeds under Assumptions 1, 2, 3, and 4, together with the boundedness result of Lemma 1.

Verification of Assumption 3. The martingale difference sequence $\{M_{m+1}\}$ arises from centered empirical averages of belief-weighted score terms together with a linear regularization term. Since the action space is finite, beliefs lie in the compact simplex Δ_K^δ, and the score function $\nabla_{\theta_k} \log L_k$ is uniformly bounded, it follows that the conditional second moment of M_{m+1} grows at most linearly in $1 + \|\Theta_m\|^2$. Thus, Assumption 3 holds.

Verification of Assumption 4. For each Θ, the set $\mathcal{G}(\Theta)$ is convex.

Proof. The set of invariant probability measures $\mathcal{I}(\Theta)$ is convex, since the invariance condition is preserved under convex combinations. Moreover, the mapping $\mu \mapsto g_\mu(\Theta)$ is linear in μ. Therefore,

$$\mathcal{G}(\Theta) = \{g_\mu(\Theta) : \mu \in \mathcal{I}(\Theta)\}$$

is convex. □

Proof of Lemma 1.

Proof. Consider the Lyapunov function

$$V(\Theta) = \|\Theta\|^2.$$

Since the action space $\mathcal{A}$ is finite and the softmax parametrization is smooth, the score function

$$\nabla_{\theta_k} \log L_k(s \mid \theta_k)$$

is uniformly bounded by some constant $C < \infty$. Hence, for any selection $\xi \in \mathcal{G}(\Theta)$,

$$\langle \Theta, \xi \rangle \leq C\|\Theta\|.$$

Because the update includes the regularization term $-\eta_{\text{reg}}\Theta$, we obtain

$$\langle \Theta, \xi - \eta_{\text{reg}}\Theta \rangle \leq C\|\Theta\| - \eta_{\text{reg}}\|\Theta\|^2.$$

For sufficiently large $\|\Theta\|$, the quadratic negative term dominates the linear term, so the drift of V is strictly negative outside a sufficiently large ball. This implies that the recursion admits a supermartingale-type bound.

More precisely, the update satisfies

$$V_{m+1} \leq V_m(1 - a_m) + b_m,$$

where $a_m = \frac{1}{2}\gamma_m \eta_{\text{reg}}$ and $b_m = C\gamma_m^2$ for some constant $C < \infty$.

Since $\sum_m a_m = \infty$ and $\sum_m b_m < \infty$, Lemma A.1 of [7] (Robbins–Siegmund) implies that

$$\sup_m \|\Theta_m\| < \infty \qquad \text{almost surely.}$$

□

Proof of Theorem 1. We map the ESL recursion to the controlled-Markov stochastic approximation framework of [6].

For fixed Θ, the fast belief process $\{B_t\}$ evolves as a time-homogeneous Markov chain on the compact state space

$$\mathcal{B} = (\Delta_K^\delta)^{N(N-1)}.$$

Since the transition kernel depends continuously on Θ, this defines a controlled Markov process in the sense of [6].

Step 1 (Noise decomposition and tracking bias). We first express the empirical recursion in stochastic approximation form by decomposing the batch update into its conditional expectation and fluctuation terms.

Let $\widehat{H}_m$ denote the empirical update computed from the m-th batch, so that

$$\Theta_{m+1} = \Theta_m + \gamma_m \widehat{H}_m. \tag{11}$$

We decompose $\widehat{H}_m$ by adding and subtracting its conditional expectation:

$$\widehat{H}_m = \xi_m + \underbrace{\big(\widehat{H}_m - \mathbb{E}[\widehat{H}_m \mid \mathcal{F}_m]\big)}_{M_{m+1}} + \underbrace{\big(\mathbb{E}[\widehat{H}_m \mid \mathcal{F}_m] - \xi_m\big)}_{\beta_m}, \tag{12}$$

where $\xi_m \in \mathcal{G}(\Theta_m)$ is a measurable selection.

Define

$$M_{m+1} := \widehat{H}_m - \mathbb{E}[\widehat{H}_m \mid \mathcal{F}_m].$$

Then $\{M_{m+1}\}$ is a martingale difference sequence since

$$\mathbb{E}[M_{m+1} \mid \mathcal{F}_m] = 0.$$

Similarly, define the tracking bias

$$\beta_m := \mathbb{E}[\widehat{H}_m \mid \mathcal{F}_m] - \xi_m.$$

Substituting into (11) gives the stochastic approximation decomposition

$$\Theta_{m+1} = \Theta_m + \gamma_m\big(\xi_m + M_{m+1} + \beta_m\big), \tag{13}$$

with drift ξ_m, martingale noise M_{m+1}, and tracking bias β_m.

The term M_{m+1} captures finite-sample fluctuations around the conditional mean, while β_m captures the discrepancy between the finite-batch conditional drift and the invariant-measure averaged drift.

Because Θ_m is held fixed within each batch, the fast belief process evolves under a frozen time-homogeneous Markov kernel on the compact state space $\mathcal{B}$. Thus, the ESL recursion falls within the controlled Markov noise framework of [6].

The decomposition above is useful for interpreting the three components of the update: ξ_m is the averaged drift, M_{m+1} is the martingale fluctuation, and β_m is the discrepancy induced by finite-batch averaging. For the convergence proof, however, we do not require a separate window-sum bound on β_m. Instead, we verify directly that the conditions of [6, Theorem 3.1] hold in our setting:

- **Boundedness:** the iterates $\{\Theta_m\}$ are almost surely bounded by Lemma 1.
- **Tightness:** the fast process evolves on the compact state space $\mathcal{B}$, so the associated occupation measures are automatically tight.
- **Kernel continuity:** the transition kernel depends on Θ through the softmax policy and the projected Bayesian update. The softmax map is smooth in Θ, and the projection onto Δ_K^δ is Lipschitz. Hence, for any bounded continuous test function f, the map

$$(B, \Theta) \mapsto \int f(B')\, P_\Theta(B, dB')$$

  is continuous.
- **Step-size conditions:** the sequence $\{\gamma_m\}$ satisfies Assumption 2.

Therefore, by Theorem 3.1 of [6], the interpolated process tracks the differential inclusion induced by invariant occupation measures of the frozen fast process. In this sense, the effect of incomplete averaging over finite batches is asymptotically negligible without requiring an explicit separate control argument for β_m.

Step 2 (Tightness of occupation measures). Following [6], consider the empirical occupation measures of the fast belief process over intervals corresponding to slow-time updates. Since the fast process evolves on the compact space

$$\mathcal{B} = (\Delta_K^\delta)^{N(N-1)},$$

the family of empirical occupation measures is automatically tight. Thus the tightness condition required in the controlled-Markov stochastic approximation framework is satisfied without any additional Foster–Lyapunov argument. Consequently, every sequence of empirical occupation measures admits weakly convergent subsequences. By the occupation-measure characterization in [6, Lemma 3.1], such limit points belong to the invariant occupation-measure set of the frozen fast process; in the present setting, since there is no auxiliary control variable beyond the frozen parameter Θ, these reduce to invariant probability measures in $\mathcal{I}(\Theta)$.

Step 3 (Asymptotic pseudotrajectory).

Interpolated trajectory. Define the cumulative time

$$T(m) := \sum_{k=0}^{m-1} \gamma_k, \quad T(0) = 0,$$

and the piecewise-linear interpolation

$$\bar{\Theta}(t) := \Theta_m + \frac{t - T(m)}{T(m+1) - T(m)}(\Theta_{m+1} - \Theta_m), \quad T(m) \le t \le T(m+1).$$

Since $\sum_m \gamma_m = \infty$, we have $T(m) \to \infty$.

Noise condition. From the decomposition in Step 1,

$$\Theta_{m+1} = \Theta_m + \gamma_m\big(\xi_m + M_{m+1} + \beta_m\big), \quad \xi_m \in \mathcal{G}(\Theta_m),$$

where M_{m+1} is a martingale difference sequence and β_m represents the tracking discrepancy arising from finite-batch averaging.

Rather than explicitly controlling β_m, we rely on the fact that the ESL recursion satisfies the controlled-Markov conditions of [6], under which the cumulative effect of the non-martingale terms is asymptotically negligible in the sense required for stochastic approximation.

The remaining stochastic fluctuations are captured by the martingale difference sequence $\{M_{m+1}\}$. Since it has bounded conditional second moments and $\sum_m \gamma_m^2 < \infty$, standard martingale arguments imply that its weighted window sums vanish almost surely. Therefore, the overall noise satisfies the standard condition required for stochastic approximation; see [3] and [4].

APT conclusion. The recursion can therefore be written in the form

$$\Theta_{m+1} = \Theta_m + \gamma_m\big(h_m + \varepsilon_m\big), \quad h_m \in \mathcal{G}(\Theta_m),$$

where $\|\varepsilon_m\| \to 0$ in the window sense.

Together with: (i) boundedness of $\{\Theta_m\}$, (ii) upper semicontinuity and compactness of $\mathcal{G}$, and (iii) the step-size conditions (including eventual monotonicity), the hypotheses of [4] are satisfied.

Therefore, the interpolated process $\bar{\Theta}(t)$ is almost surely an asymptotic pseudotrajectory of the differential inclusion

$$\dot{\Theta} \in \mathcal{G}(\Theta),$$

i.e., for every $T > 0$,

$$\lim_{\tau\to\infty} \sup_{s\in[\tau,\tau+T]} d\big(\bar{\Theta}(s), \mathcal{G}(\bar{\Theta}(s))\big) = 0 \quad \text{a.s.}$$

Step 4 (Limit set characterization).

Since $\{\Theta_m\}$ is almost surely bounded, the interpolated trajectory $\bar{\Theta}(t)$ is a bounded asymptotic pseudotrajectory of the differential inclusion

$$\dot{\Theta} \in \mathcal{G}(\Theta). \tag{14}$$

By standard results on asymptotic pseudotrajectories for differential inclusions [3, 4], the limit set of $\bar{\Theta}(t)$, and hence of $\{\Theta_m\}$, is almost surely a compact internally chain transitive (ICT) set of the differential inclusion.

This proves Theorem 1.

Proof of Corollary.

Proof. If $\mathcal{I}(\Theta)$ is a singleton for each Θ, then the set-valued mean-field map $\mathcal{G}(\Theta)$ reduces to a single-valued vector field $g(\Theta)$, so the differential inclusion (14) collapses to the ODE

$$\dot{\Theta} = g(\Theta). \tag{15}$$

Under the stated global asymptotic stability assumption, Θ^* is an attractor whose basin contains the almost-sure limit set of the iterates. By Theorem 1, the limit set of $\{\Theta_m\}$ is almost surely an internally chain transitive (ICT) set of the ODE (15). By [3, Corollary 5.4], any ICT set meeting the basin of an attractor is contained in that attractor. Therefore the limit set is contained in $\{\Theta^*\}$, and hence

$$\Theta_m \to \Theta^* \qquad \text{almost surely.}$$

□

A.1 Interpretation of ICT Limit Sets

Theorem 1 characterizes the limiting behavior of ESL through internally chain transitive (ICT) sets of the differential inclusion (14). In the present setting, these ICT sets correspond to self-consistent regimes of the endogenous belief–policy–data feedback loop: for a given Θ, the fast belief dynamics induce invariant distributions over interaction states; these invariant distributions determine averaged update directions in $\mathcal{G}(\Theta)$; and the resulting prototype parameters remain consistent with the data distribution generated by the induced policies.

Thus, ICT sets serve as generalized equilibrium objects for ESL under endogenous, potentially non-ergodic dynamics. When the invariant measure is unique, the set-valued dynamics collapse to an ordinary differential equation, in which case the ICT limit sets coincide with invariant sets of the associated ODE, and reduce to fixed points under additional stability conditions.

B Additional Experimental Details and Analysis

This appendix provides additional experimental details, extended analyses, and robustness evaluations complementing the results in the main paper.

B.1 Implementation Details

Training setup. All methods are trained using identical policy parameterizations and optimization settings where applicable. Policies are parameterized as softmax distributions over action logits. Gradients are estimated using minibatch updates over interaction data, with learning rates and optimization schedules matched across methods where possible.

Hyperparameters. Unless otherwise specified, experiments use:

- Number of prototypes: $K = 5$
- Batch size: $M = 50$
- Regularization coefficient: $\eta_{\text{reg}} =$ [placeholder]
- Inverse temperature: $\lambda =$ [placeholder]

Evaluation protocol. All experiments are run over multiple random seeds, and results are reported with mean and 95% confidence intervals.

B.2 Robustness to Observation Noise and Partial Observability

We evaluate ESL under perturbations to the observation process to assess robustness beyond the idealized noiseless setting.

Setup. We introduce:

- **Observation noise:** stochastic corruption of observed actions,
- **Partial observability:** reduced probability of observing opponent actions.

[Placeholder: Robustness plots]

Analysis. ESL maintains stable performance under moderate noise and partial observability, indicating that belief-based inference remains effective under imperfect signals.

B.3 Belief Dynamics and Entropy

We analyze the evolution of belief distributions $b_{i \to j}$ over time.

Metrics. We track:

- Average belief entropy,
- Distribution of belief concentration across agents.

[Placeholder: Belief entropy plots]

Analysis. Beliefs progressively concentrate as agents accumulate interaction data, reflecting increasing certainty over latent behavioral types. Under higher non-stationarity, entropy stabilizes at intermediate levels, indicating persistent uncertainty. These trends are consistent with the role of beliefs as fast-timescale variables adapting under endogenous interaction dynamics.

B.4 Prototype Structure and Separation

We examine the structure of learned prototypes over training.

Metrics. We measure pairwise distances between prototype parameters and track their evolution over time.

[Placeholder: Prototype separation plots]

Analysis. Distinct prototypes emerge and stabilize during training, indicating that ESL learns a structured latent representation rather than collapsing to degenerate solutions.

B.5 Additional Ablations

Effect of number of prototypes K. We evaluate ESL under varying K relative to the true number of latent behavioral types.

Setup. We fix the true number of types and vary $K \in \{1, 2, 3, 5, 10\}$.

[Placeholder: Performance vs K plots]

Analysis. When K is under-specified, ESL fails to capture heterogeneity and performance degrades. When over-specified, redundant prototypes emerge but performance degrades gracefully, indicating robustness to misspecification.

Extended ablations. We report additional results for:

- No belief update,
- No shared prototypes,
- No epistemic weighting,
- No projection (removal of projection constraints leading to instability in long-run updates).

[Placeholder: Extended ablation tables/plots]

B.6 Additional Dynamical Visualizations

We provide extended visualizations of prototype trajectories and belief evolution across different environments.

[Placeholder: Additional trajectory plots]

Analysis. These visualizations further illustrate the range of dynamical behaviors observed in ESL, including convergence to stable configurations and persistent oscillations under non-stationary interaction regimes.